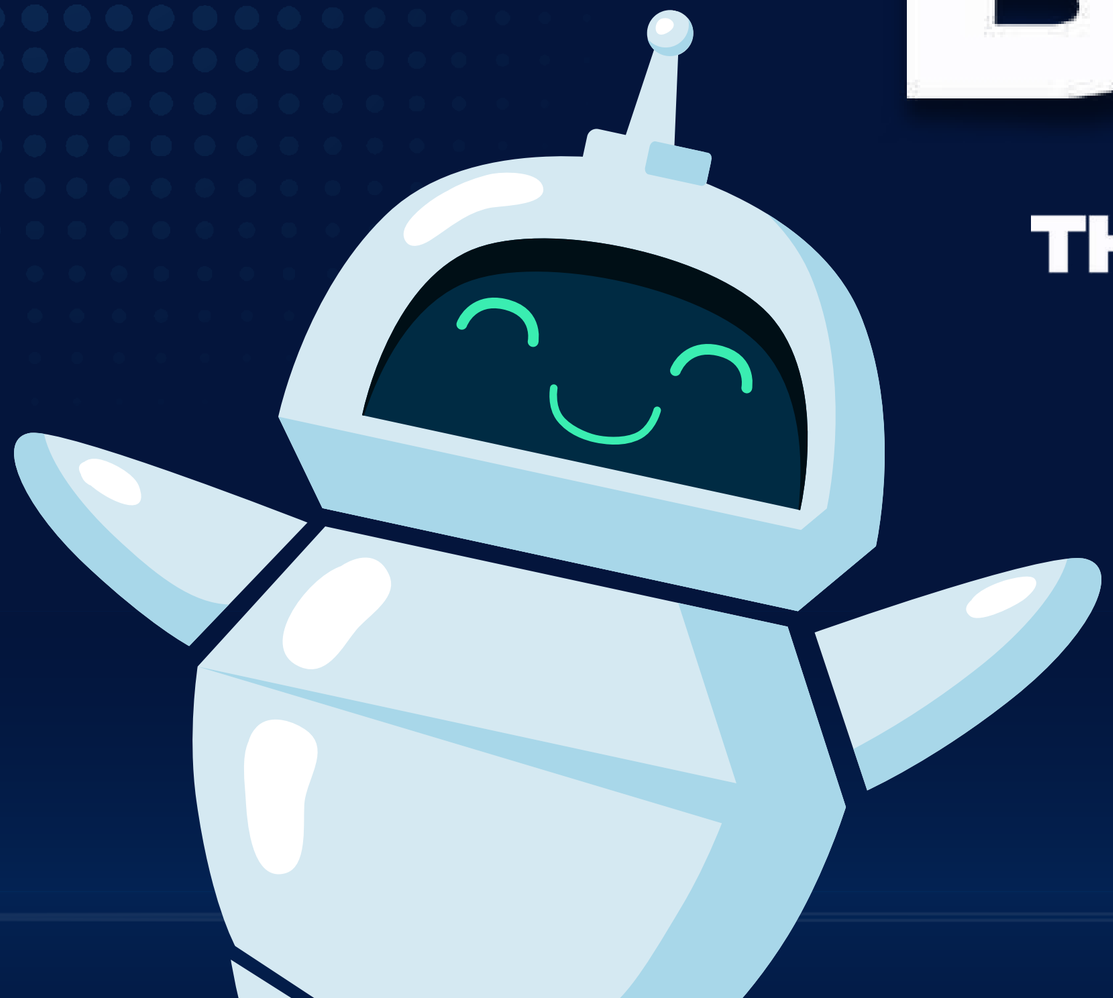
BUDDY

THE EMOTIONAL SUPPORT CHATBOT



INTRODUCING BUDDY

EMOTIONAL SUPPORT AND EMPATHY

- an AI-powered chatbot meticulously designed to provide emotional comfort and empathy when users need it most
- actively listens and responds with a human like, profound understanding of your feelings, creating a safe, non-judgmental space
- helps users regain calm and clarity, a grounding presence during moments of emotional distress.

YOUR OWN EMOTIONAL SUPPORT CHATBOT

MORE THAN A BOT

- a compassionate companion available at your fingertips, ready to support your mental well-being

KEY FEATURES

1 MULTIPLE INTERACTION MODES

- Normal Mode: Everyday emotional assistance.
- Hands-Free Mode: Natural, flowing, friend like conversations.
- Private Mode: Safe space for sensitive or deeply personal discussions.

2 INTELLIGENT PERSONALISATION

System tracks user data and emotional patterns in an SQLite3 database, allowing BUDDY to provide meaningful follow-up questions and tailored support in future sessions.

3 HANDS FREE CONVERSATIONS

Hands-Free mode makes interactions feel lively and spontaneous, giving users the sense of a natural, flowing conversation with a friend.

4 REALISM AND PRIVACY

Private mode ensures confidentiality and user trust by erasing session records once closed, while still offering realistic, empathetic conversations.

SPEECH INTERACTION

- The chatbot uses Vosk-model-small-en-us-0.15 to convert spoken words into accurate text, ensuring that the system understands user input naturally.
- It processes the audio in real time with SoundDevice, maintaining a smooth flow without delays.
- Finally, it uses Edge-TTS to turn responses into clear, natural-sounding speech, creating a seamless two-way conversation.

EMOTION ANALYSIS

- Uses TextBlob to analyze the sentiment, polarity categorizes crisis help and detects emotions such as happiness, sadness, or frustration and lot more.

- Employs a custom emotion lexicon to interpret the detected emotions in greater depth. The lexicon helps to categorise the emotion in detail and form specific instructions based on it.

- Generates prompts using the lexicon to create specific conversational rules: what to say, what to avoid, and how to set the tone.

SMART RESPONSES

- Once the chatbot understands the user's emotions and intent, using the specified tailored prompt, it begins generating a reply.
- Uses the OpenRouter API to craft context-aware and emotion-sensitive responses.
- Takes into account both the current mood and the conversation history using database entries, unlike generic chatbots.
- Delivers personalized, human-like dialogue that feels natural and meaningful instead of scripted.

HARDWARE AND UI

- The UI is still under development but our approach is for a light colored, stress relieving UI with faces.
- It is powered through a 5V/3A power adapter which handles the pi, a waveshare 7 inch capacitive touch display, a USB microphone and a set of compatible speakers.
- The terminal OS releases a bit more ram for our bot and we trigger auto run of the bot making it look like a standalone device.
- All this is masked over a raspberry pi OS lite running on raspberry pi3b+ with 1gb LPDDR2 ram and usb/3.5mm jack support of microphone, speaker and keyboard.

EFFICIENCY

- Behind the scenes, the chatbot uses queues, threads, and synchronization to handle tasks smoothly.
- These techniques ensure audio input and output are managed without delays or interruptions.
- This prevents lags, overlaps, or glitches that could disrupt the flow of conversation.

Here are the source files for the bot :-

BUDDY

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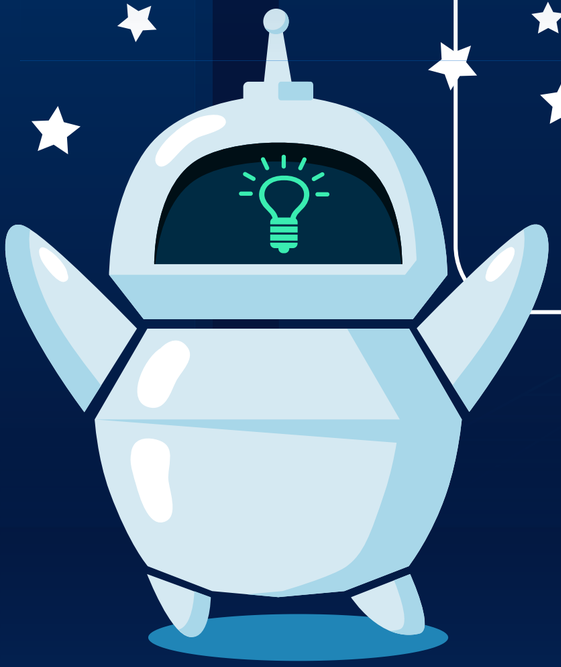

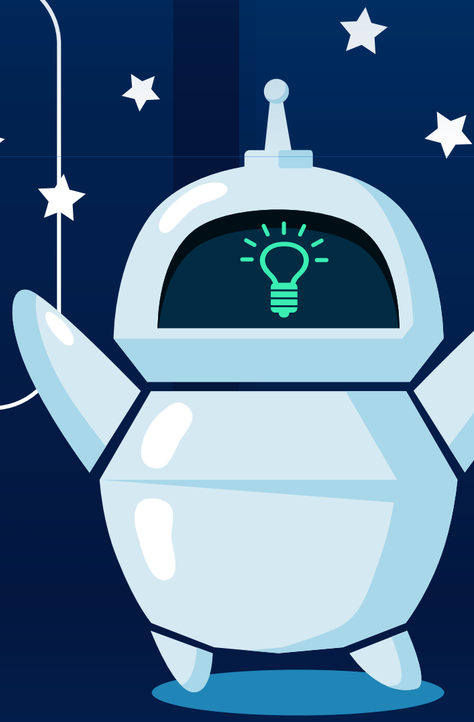

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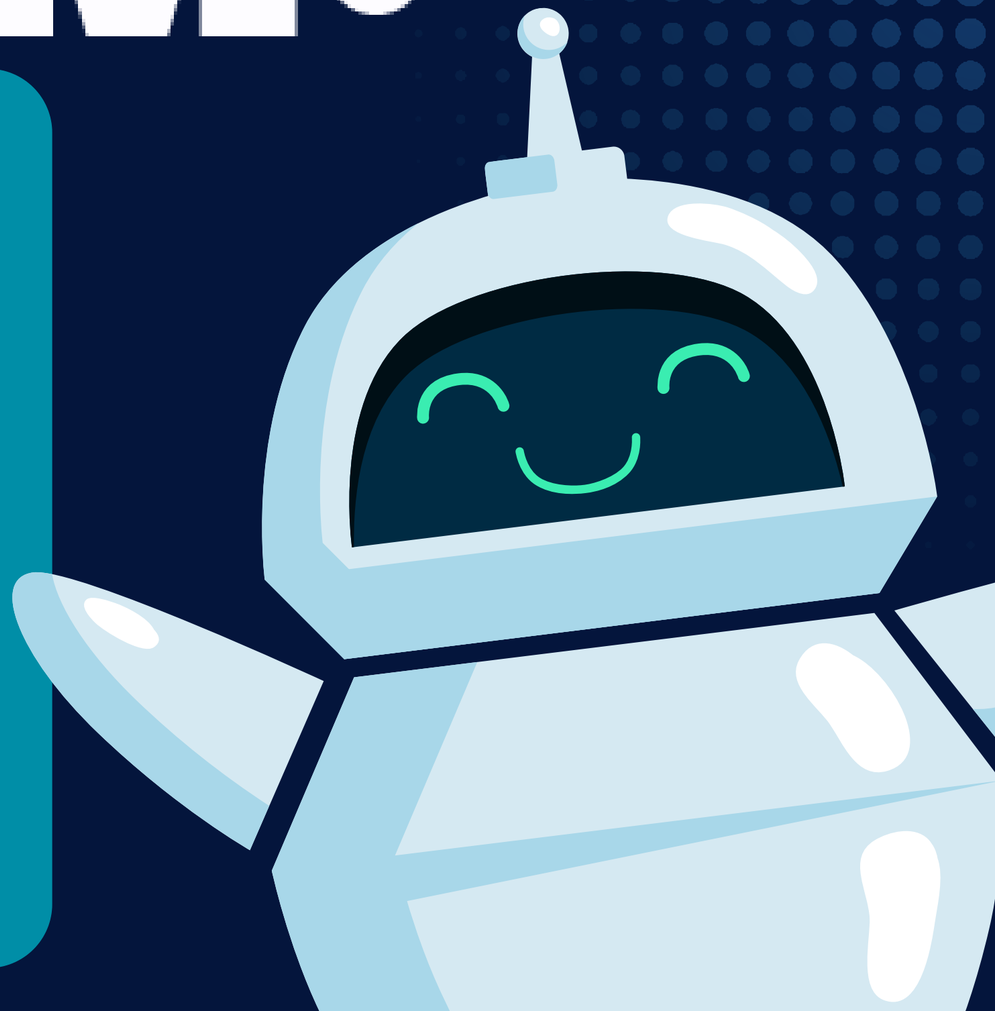
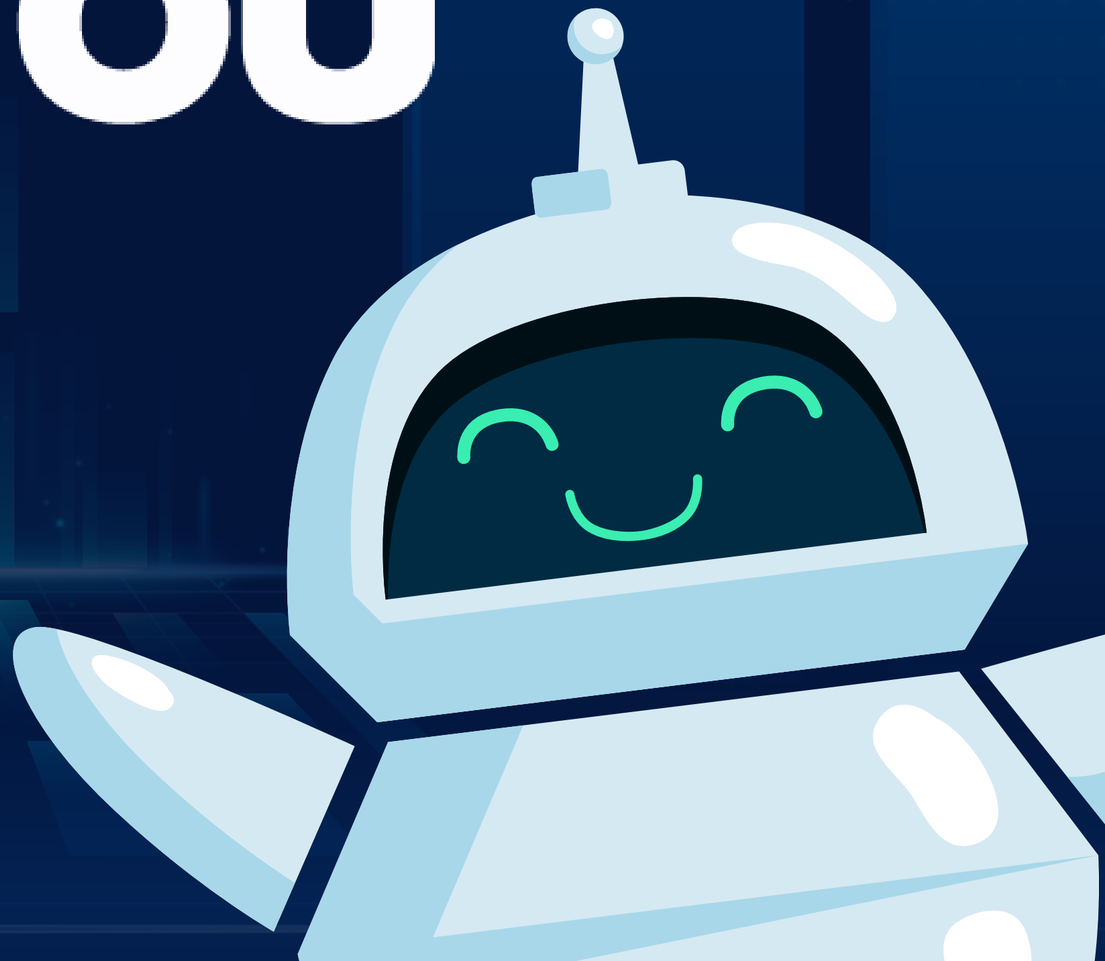
THANK YOU